Experiment 12: Conduct Z test for the Data Given using Python Objective

Code:

```
import numpy as np
from math import sqrt
from scipy import stats
from scipy.stats import norm

print("=== EXP NO: 12: Z-TEST, t-TEST, F-TEST ANOVA ===\n")

# 1. One-Sample t-Test[span_2](start_span)[span_2](end_span)
print("--- 1. One-Sample t-Test ---")
marks = np.array([82, 79, 85, 88, 76, 81, 84, 80, 86, 83])
mu_0_t = 86
t_stat, p_value_t = stats.ttest_1samp(marks, mu_0_t)

print(f"T-Statistic: {t_stat:.3f}")
print(f"P-value: {p_value_t:.4f}")

alpha = 0.05
if p_value_t < alpha:
    print("Result: Reject Null Hypothesis\n")
else:
    print("Result: Fail to Reject Null Hypothesis\n")
```

```
# 2. Z-Test[span_3](start_span)[span_3](end_span)
print("--- 2. Z-Test ---")
x_bar = 49.5
mu_0_z = 50
sigma = 4
n = 64

z_stat = (x_bar - mu_0_z) / (sigma / sqrt(n))
p_value_z = 2 * (1 - norm.cdf(abs(z_stat)))

print(f"Z-statistics: {z_stat:.3f}")
print(f"P-value: {p_value_z:.4f}")

if p_value_z < alpha:
    print("Result: Reject Null Hypothesis\n")
else:
    print("Result: Fail to Reject Null Hypothesis\n")



# 3. One-Way ANOVA (F-
Test)[span_4](start_span)[span_4](end_span)
print("--- 3. One-Way ANOVA (F-Test) ---")
```

```
A = [15, 18, 17]

B = [22, 24, 23]

C = [19, 21, 20]


f_stat, p_value_f = stats.f_oneway(A, B, C)


print(f"F-statistics: {f_stat:.3f}")

print(f"P-value: {p_value_f:.4f}")


if p_value_f < alpha:

   print("Result: Reject Null Hypothesis")

else:

   print("Result: Fail to Reject Null Hypothesis")
```

Output:

```
Sample Mean: 150.20

Z-Statistic: 0.6406

P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.
```

Result:

Thus the experiment was executed successfully and Z test for the Data Given using Python Objective was analysed.